

Chapter 5: The Structure and Function of Large Biological Molecules

Complete Course Curriculum Master Reference • Verbatim Textbook Edition

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Module 0: Chapter 5 Content Introduction

Overview of Macromolecules, Monomers vs. Polymers, and Condensation / Hydrolysis Reactions

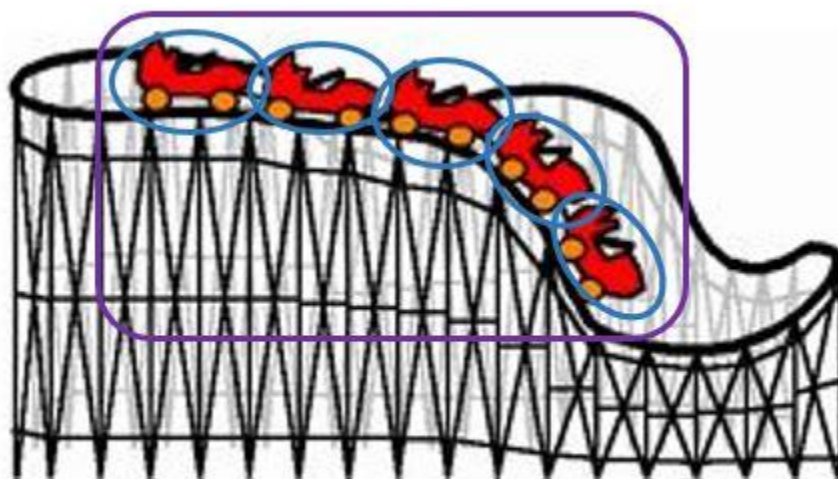
Chapter 5 Content Introduction

Learning Objectives

- Understand that macromolecules are polymers, constructed from small monomers
- Synthesize polymers from monomers, breakdown polymers into monomers
- List the 4 types of biological macromolecules (Carbohydrate, lipids, proteins, nucleic acids)
- Recognize that carbohydrates provide energy for the cell
- Recognize the lipids are fats and steroids
- Recognize that proteins are diverse structures with a wide range of cellular functions
- Recognize that nucleic acids store and transmit genetic information

Chapter 5: The Structure and Function of Large Biological Molecules

There are four types of macromolecules that are biologically important: Carbohydrates, Lipids, Proteins, and Nucleic Acids. Three of these four macromolecules (carbohydrates, proteins, and nucleic acids) are **polymers**; one long molecule constructed by linking many, smaller **monomers** together via covalent bonds. Roller coasters make a great analogy for this: the roller coaster (the polymer) is a long train constructed by linking smaller, individual cars (the monomers) together. Lipids are macromolecules but they are not true polymers. The tricky part is understanding how to synthesize (construct) polymers from monomers and how to break polymers down into individual monomers.

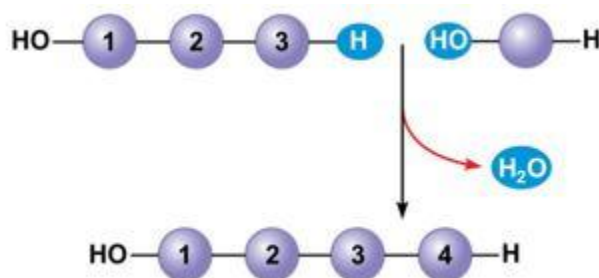


The **roller coaster** is the polymer and the individual **cars** are the monomers

Interactive Course Widget: Polymers and Monomers widget

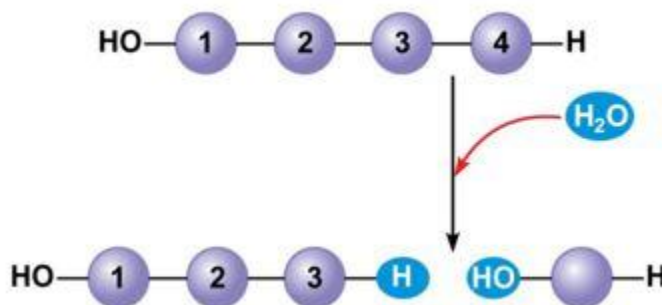
Cells synthesize (build) polymers from monomers via condensation reactions, and cells disassemble polymers into individual monomers via hydrolysis reactions. As you might have guessed from the names, both reactions involve water; condensation creates a water molecule and hydrolysis splits a water molecule.

The chemistry for each type of monomer is essentially the same; there is a hydrogen (H) on one side of the monomer and a hydroxide (OH) on the other side. When two monomers are linked together, the hydrogen from one molecule is removed and the hydroxide from the other monomer is removed; this creates a covalent bond between the monomers and the H and OH bond to each other to create a water molecule. Condensation reactions require energy.



So, let's take a look at the polymer shown above. This polymer is the result of linking four monomers. First, a bond was created between monomer 1 and monomer 2, the creation of this bond released one water molecule. Then a bond was created between the 1--2 polymer and monomer 3, the creation of this bond released one water molecule. Last, a bond was created between the 1--2--3 polymer and monomer 4, the creation of this bond released one water molecule. So, putting it all together, linking four monomers into one polymer released three water molecules. The number of water molecules released is one less than the number of monomers linked together to create the polymer. If you link six monomers to create one polymer, how many water molecules are released?

When a polymer is disassembled into individual monomers, the covalent bond between them is broken. When a covalent bond is broken, it must be replaced so that the valence shells of each monomer are filled. A water molecule is split into one hydrogen (H) and one hydroxide (OH); the hydrogen bonds to one monomer and the OH bonds to the other monomer. Hydrolysis reactions release energy.



If we look at the polymer shown above. We can disassemble this polymer into four individual monomers. First, let's look at removing just the 4 monomer from 1--2--3--4 polymer. To remove the 4 monomer, we must break the covalent bond between monomer 3 and monomer 4. Breaking this covalent bond results in both monomers missing electrons in their outer shells. Splitting a water molecule will allow the hydrogen (H) atom to bond to the remaining 1--2--3 polymer and hydroxide (OH) will bond to the 4 monomer. Next, we need to remove the 3 monomer from the 1--2--3 polymer by again splitting a water molecule; the hydrogen atom will bond with the resulting 1--2 polymer and the hydroxide will bond to the 3 monomer. Last, we will split the 1--2 polymer into two individual monomers by again splitting one water molecule so that the hydrogen bonds with the 1 monomer and the hydroxide bonds with the 2 monomer. So, putting it all together, completely disassembling a polymer comprised of four monomers into individual monomers requires three water molecules. The number of required water molecules is one less than the number of monomers linked together to create the polymer. If you disassemble a polymer comprised of eight monomers, how many water molecules are required?

[Interactive Course Widget: Condensation and Dehydration Reaction widget](#)

CHAPTER READING

Chapter 5 [Carbohydrates](#), [Proteins](#), [Nucleic Acids](#), [Lipids](#)

CHAPTER LINKS

Chapter 5 PowerPoint: [Part 1](#); [Part 2](#)

[Chapter 5 Simple Practice Problems](#)

[Chapter 5 Post-Quiz](#)

Module 1: Chapter 5 Carbohydrates

Monosaccharides, Disaccharides, Polysaccharides, and Glycosidic Linkages

Chapter 5 Carbohydrates

Carbohydrates

Learning Objectives

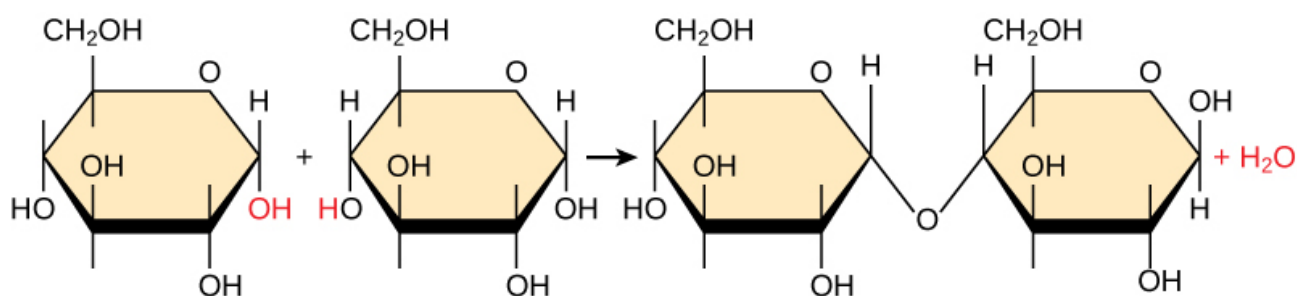
By the end of this section, you will be able to do the following:

- Understand macromolecule synthesis
- Explain dehydration (or condensation) and hydrolysis reactions
- Discuss the role of carbohydrates in cells and in the extracellular materials of animals and plants
- Explain carbohydrate classifications

As you've learned, biological macromolecules are large molecules, necessary for life, that are built from smaller organic molecules. There are four major biological macromolecule classes (carbohydrates, lipids, proteins, and nucleic acids). Each is an important cell component and performs a wide array of functions. Combined, these molecules make up the majority of a cell's dry mass (recall that water makes up the majority of its complete mass). Biological macromolecules are organic, meaning they contain carbon. In addition, they may contain hydrogen, oxygen, nitrogen, and additional minor elements.

Condensation (Dehydration) Synthesis

Most macromolecules are made from single subunits, or building blocks, called monomers. The monomers combine with each other using covalent bonds to form larger molecules known as polymers. In doing so, monomers release water molecules as byproducts. This type of reaction is dehydration synthesis, which means "to put together while losing water."



Shown is the reaction of two glucose monomers to form maltose. When maltose is formed, a water molecule is released. The components of the linkage are upper case O upper case H from one glucose molecule combining with o...

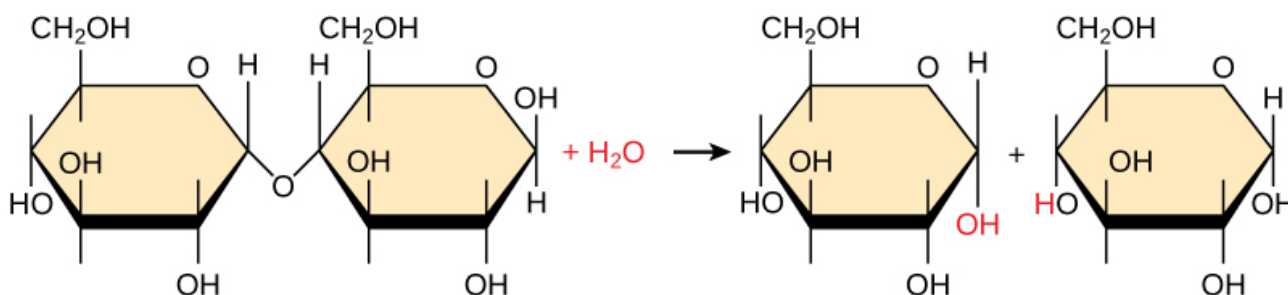
Figure 3.2 In the dehydration synthesis reaction above, two glucose molecules link to form the disaccharide maltose. In the process, it forms a water molecule.

In a dehydration synthesis reaction ([Figure 3.2](#)), the hydrogen of one monomer combines with the hydroxyl group of another monomer, releasing a water molecule. At the same time, the monomers share electrons and form covalent bonds. As additional monomers join, this chain of repeating monomers forms a polymer. Different

monomer types can combine in many configurations, giving rise to a diverse group of macromolecules. Even one kind of monomer can combine in a variety of ways to form several different polymers. For example, glucose monomers are the constituents of starch, glycogen, and cellulose.

Hydrolysis

Polymers break down into monomers during hydrolysis. A chemical reaction occurs when inserting a water molecule across the bond. Breaking a covalent bond with this water molecule in the compound achieves this (Figure 3.3). During these reactions, the polymer breaks into two components: one part gains a hydrogen atom (H^+) and the other gains a hydroxyl molecule (OH^-) from a split water molecule.



Shown is the breakdown of maltose to form two glucose monomers. Water is a reactant. The water molecule, upper case H subscript 2 baseline upper case O, breaks apart, with upper O upper H obtained by one of the glucose ...

Figure 3.3 In the hydrolysis reaction here, the disaccharide maltose breaks down to form two glucose monomers by adding a water molecule. Note that this reaction is the reverse of the synthesis reaction in Figure 3.2.

Most people are familiar with carbohydrates, one type of macromolecule, especially when it comes to what we eat. To lose weight, some individuals adhere to “low-carb” diets. Athletes, in contrast, often “carb-load” before important competitions to ensure that they have enough energy to compete at a high level. Carbohydrates are, in fact, an essential part of our diet. Grains, fruits, and vegetables are all natural carbohydrate sources that provide energy to the body, particularly through glucose, a simple sugar that is a component of starch and an ingredient in many staple foods. Carbohydrates also have other important functions in humans, animals, and plants.

Monosaccharides

Monosaccharides (mono- = “one”; sacchar- = “sweet”) are simple sugars, the most common of which is glucose. In monosaccharides, the number of carbons is typically 5-6.

The chemical formula for glucose is $C_6H_{12}O_6$. In humans, glucose is an important source of energy. During cellular respiration, energy releases from glucose, and that energy helps make adenosine triphosphate (ATP). Plants synthesize glucose using carbon dioxide and water, and glucose in turn provides energy requirements for the plant.

Galactose (part of lactose, or milk sugar) and fructose (found in sucrose, in fruit) are other common monosaccharides. Although glucose, galactose, and fructose all have the same chemical formula ($C_6H_{12}O_6$), they differ structurally and chemically (and are isomers) because of the different arrangement of functional groups around the asymmetric carbon.

VISUAL CONNECTION

Visual Connection

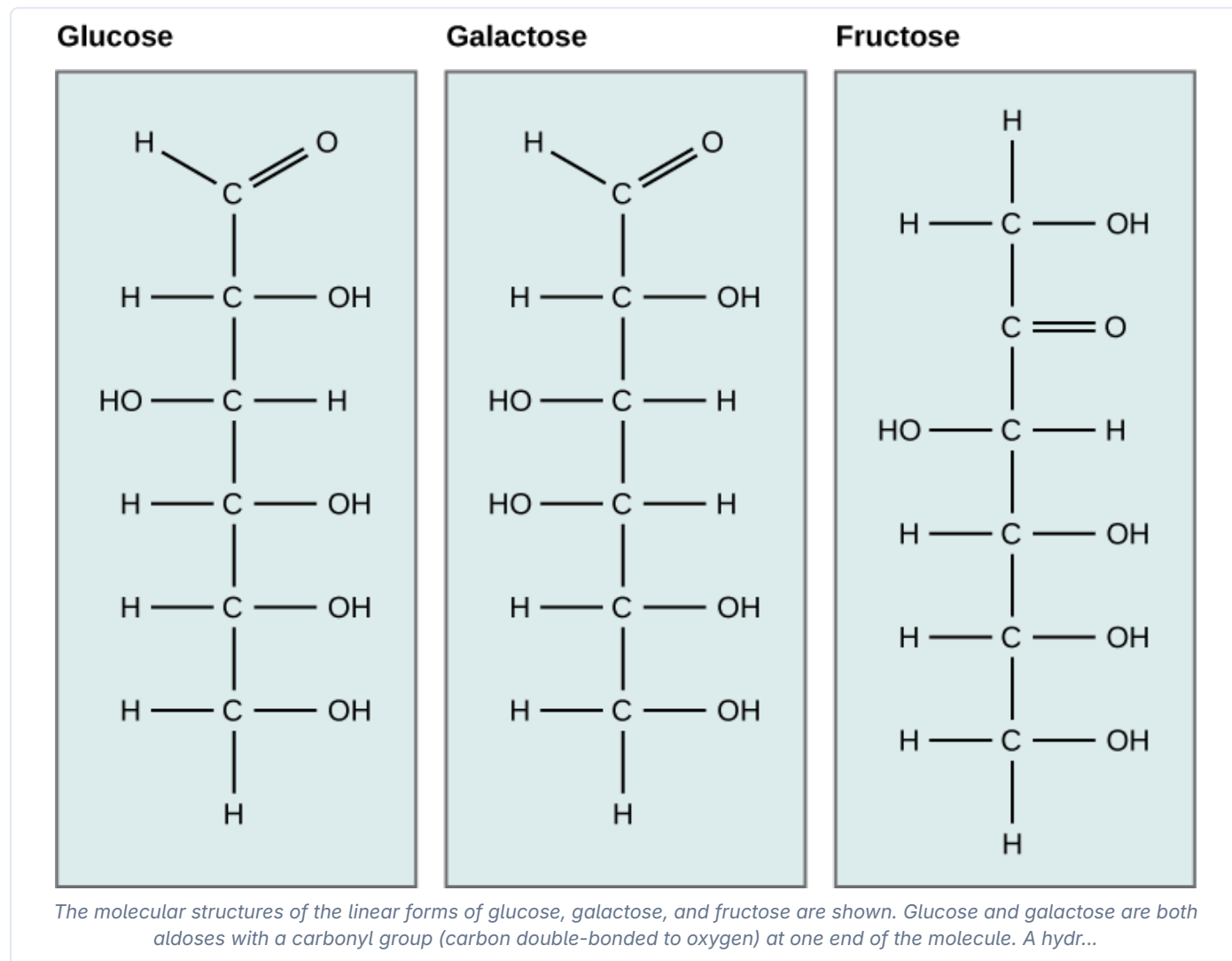
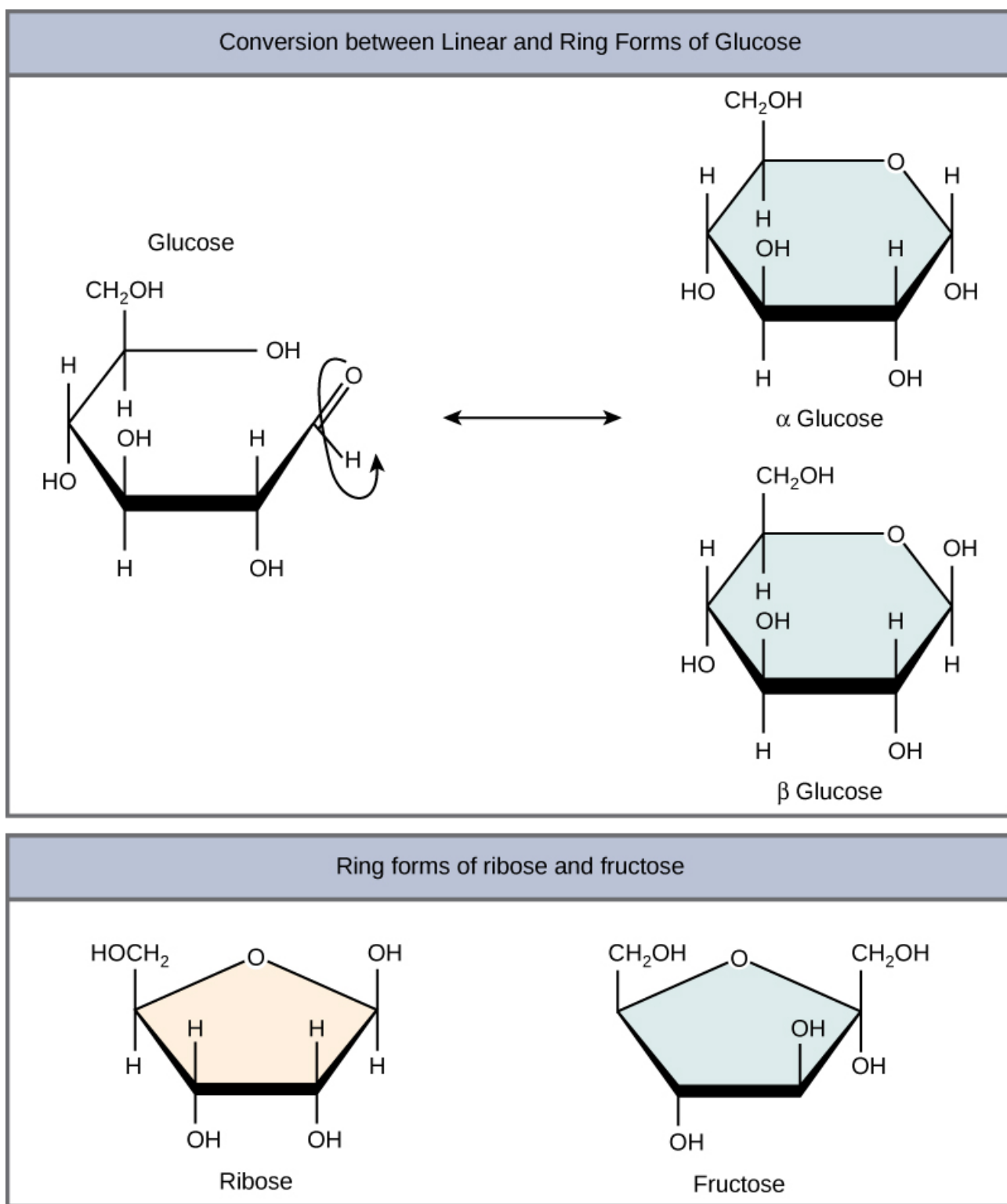


Figure 3.5 Glucose, galactose, and fructose are all hexoses. They are structural isomers, meaning they have the same chemical formula ($\text{C}_6\text{H}_{12}\text{O}_6$) but a different atom arrangement.

Monosaccharides can exist as a linear chain or as ring-shaped molecules. In aqueous solutions they are usually in ring forms (Figure 3.6). Glucose in a ring form can have two different hydroxyl group arrangements (OH) around the anomeric carbon (carbon 1 that becomes asymmetric in the ring formation process). If the hydroxyl group is below carbon number 1 in the sugar, it is in the alpha (α) position, and if it is above the plane, it is in the beta (β) position.

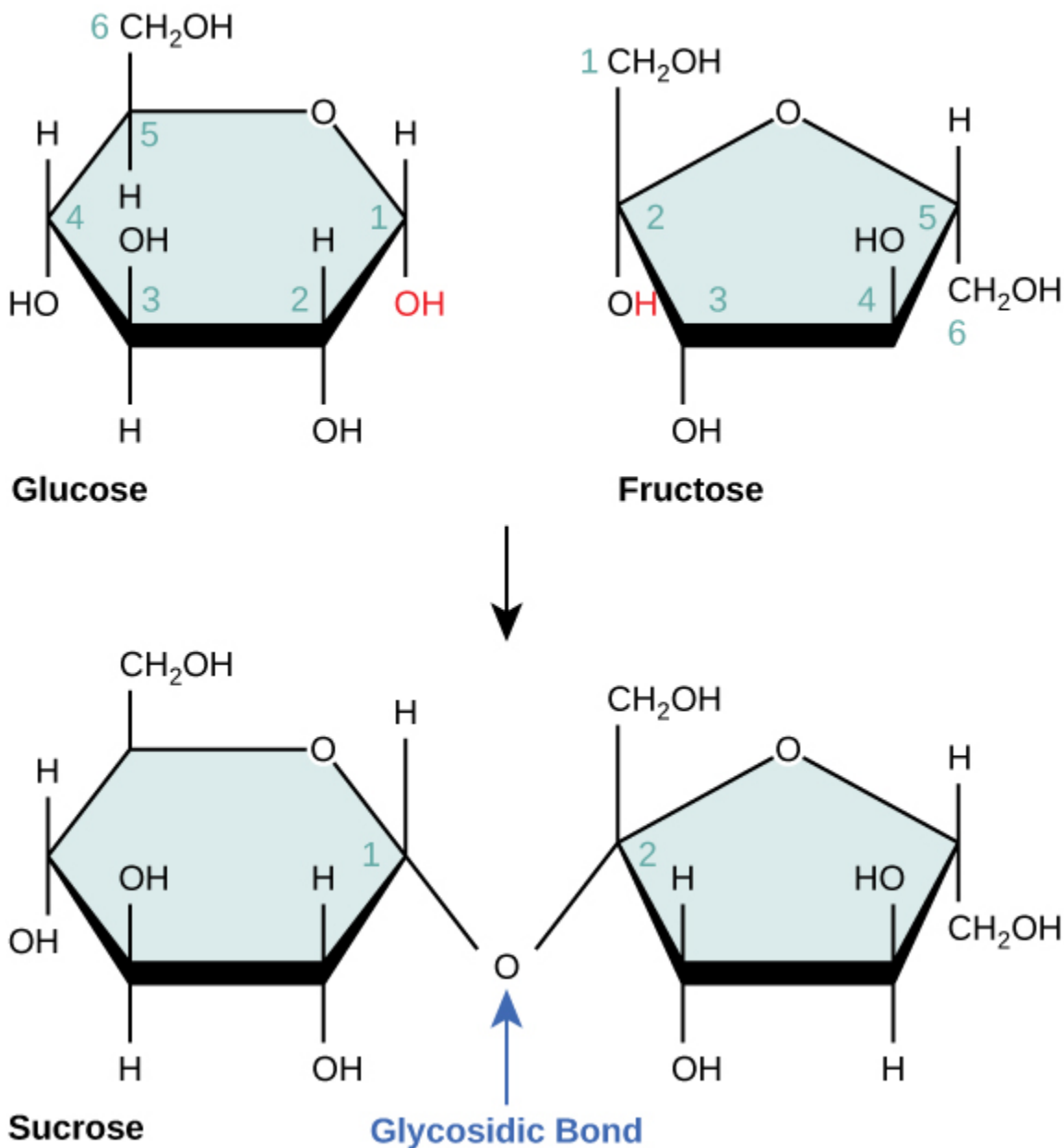


The conversion of glucose between linear and ring forms is shown. The glucose ring has five carbons and an oxygen. In alpha glucose, the first hydroxyl group is locked in a down position, and in beta glucose, the ring is...

Figure 3.6 Five and six carbon monosaccharides exist in equilibrium between linear and ring forms. When the ring forms, the side chain it closes on locks into an α or β position. Fructose and ribose also form rings, although they form five-membered rings as opposed to the six-membered ring of glucose.

Disaccharides

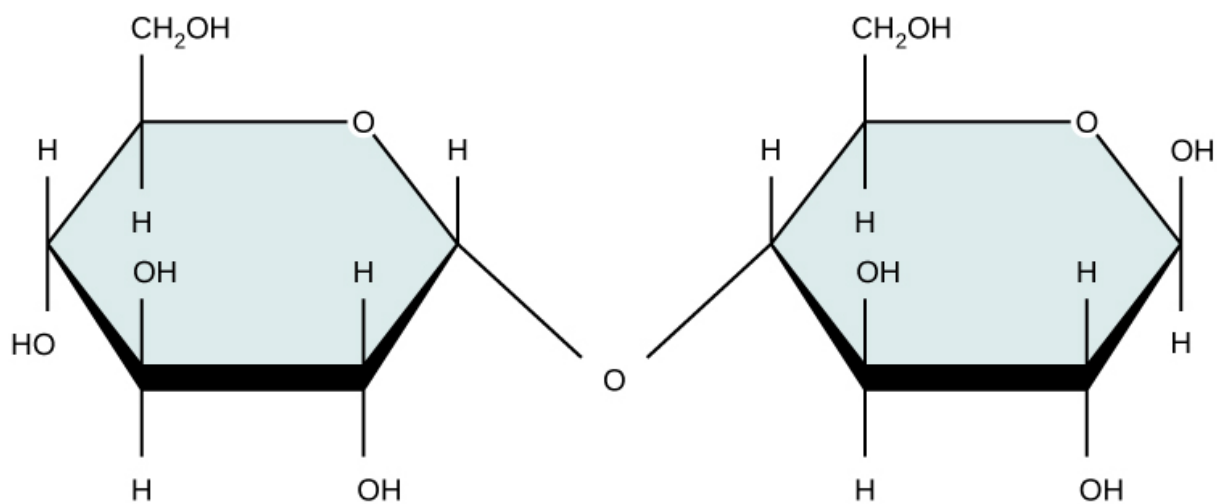
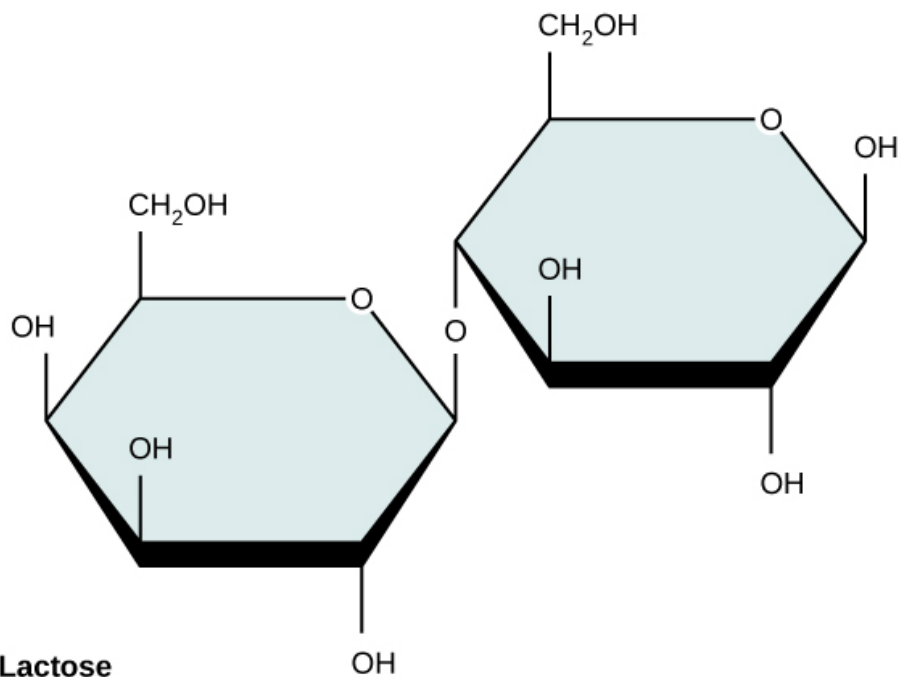
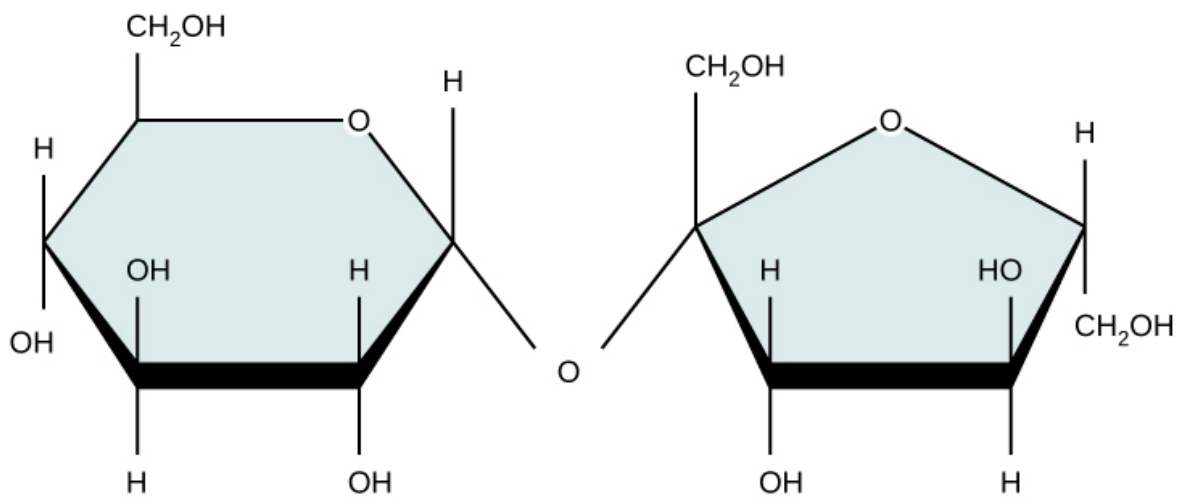
Disaccharides (di- = "two") form when two monosaccharides undergo a condensation reaction. During this process, one monosaccharide's hydroxyl group combines with another monosaccharide's hydrogen, releasing a water molecule and forming a covalent bond. A covalent bond, called a glycosidic linkage, forms between the two monosaccharides.



The formation of sucrose from glucose and fructose is shown. In sucrose, the number one carbon of the glucose ring is connected to the number two carbon of fructose via an oxygen.

Figure 3.7 Sucrose forms when a glucose monomer and a fructose monomer join in a dehydration reaction to form a glycosidic bond. In the process, a water molecule is lost. By convention, the carbon atoms in a monosaccharide are numbered from the terminal carbon closest to the carbonyl group. In sucrose, a glycosidic linkage forms between carbon 1 in glucose and carbon 2 in fructose.

Common disaccharides include lactose, maltose, and sucrose ([Figure 3.8](#)). Lactose is a disaccharide consisting of the monomers glucose and galactose. It is naturally in milk. Maltose, or malt sugar, is a disaccharide formed by a dehydration reaction between two glucose molecules. The most common disaccharide is sucrose, or table sugar, which is comprised of glucose and fructose monomers.

**Maltose****Lactose****Sucrose**

The chemical structures of maltose, lactose, and sucrose are shown. Both maltose and lactose are made from two glucose monomers joined together in ring form. In maltose, the oxygen in the glycosidic bond points downward....

Figure 3.8 Common disaccharides include maltose (grain sugar), lactose (milk sugar), and sucrose (table sugar).

Polysaccharides

A long chain of monosaccharides linked by glycosidic bonds is a polysaccharide (poly- = "many"). The chain may be branched or unbranched, and it may contain different types of monosaccharides. Starch, glycogen, cellulose, and chitin are primary examples of polysaccharides.

Plants store carbohydrates in the form of starch and cellulose. Plants are able to synthesize glucose, and they store the excess glucose, beyond their immediate energy needs, as starch in different plant parts, including roots and seeds. The starch in the seeds provides food for the embryo as it germinates and can also act as a food source for humans and animals. Cellulose mostly comprises a plant's cell wall. This provides the cell structural support. Wood and paper are mostly cellulosic in nature.

Glycogen is the storage form of glucose in humans and other vertebrates and is comprised of monomers of glucose. Glycogen is the animal equivalent of starch and is a highly branched molecule usually stored in liver and muscle cells. Whenever blood glucose levels decrease, glycogen breaks down to release glucose in a process scientists call glycogenolysis.

Module 2: Chapter 5 Lipids

Triacylglycerols, Saturated vs. Unsaturated Fatty Acids, Phospholipids, and Steroids

Chapter 5 Lipids

Learning Objectives

By the end of this section, you will be able to do the following:

- Describe the three major types of lipids
- Differentiate between saturated and unsaturated fatty acids
- Describe phospholipids and their role in cells
- Define the basic structure of a steroid and some steroid functions

Lipids include a diverse group of compounds that are largely nonpolar in nature. This is because they are hydrocarbons that include mostly nonpolar carbon–carbon or carbon–hydrogen bonds. Non-polar molecules are hydrophobic (“water fearing”), or insoluble in water. Lipids perform many different functions in a cell. Cells store energy for long-term use in the form of fats. Lipids also provide insulation from the environment for plants and animals ([Figure 3.12](#)). For example, they help keep aquatic birds and mammals dry when forming a protective layer over fur or feathers because of their water-repellant hydrophobic nature. Lipids are also the building blocks of many hormones and are an important constituent of all cellular membranes. Lipids include fats, oils, waxes, phospholipids, and steroids.



Photo shows a river otter swimming. The river otter has thick fur that is smooth and wet.

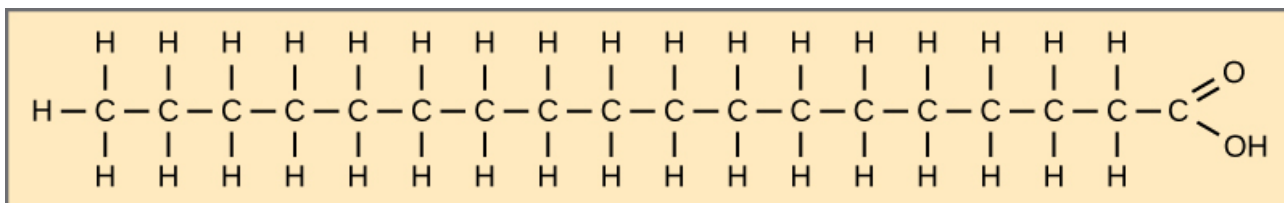
Figure 3.12 Hydrophobic lipids in aquatic mammals' fur, such as this river otter, protect them from the elements. (credit: Ken Bosma)

Fats

A fat molecule consists of two main components—glycerol and fatty acids. Glycerol is an organic compound with three carbons, five hydrogens, and three hydroxyl (OH) groups. Fatty acids have a long chain of hydrocarbons to which a carboxyl group is attached, hence the name “fatty acid.” In a fat molecule, the fatty acids attach to each of the glycerol molecule's three carbons with an **ester bond** through an oxygen atom ([Figure 3.13](#)).

Figure 3.13 Joining three fatty acids to a glycerol backbone in a dehydration reaction forms triacylglycerol. Three water molecules release in the process.

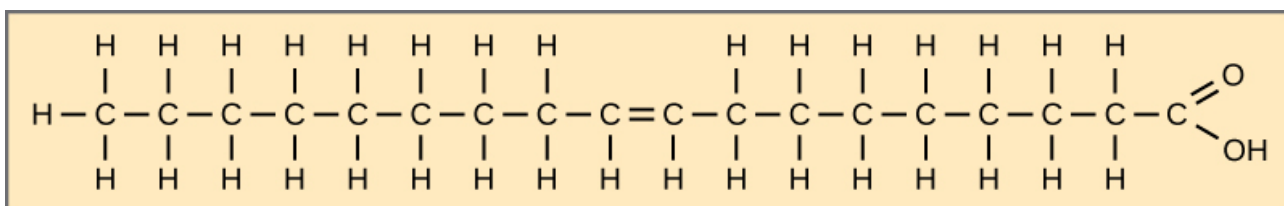
The three fatty acids in the triacylglycerol may be similar or dissimilar. We also call fats triglycerides because of their chemical structure. Fatty acids may be saturated or unsaturated. In a fatty acid chain, if there are only single bonds between neighboring carbons in the hydrocarbon chain, the fatty acid is saturated. Saturated fatty acids are saturated with hydrogen. In other words, the number of hydrogen atoms attached to the carbon skeleton is maximized. When a fatty acid has no double bonds, it is a saturated fatty acid because it is not possible to add more hydrogen to the chain's carbon atoms. Animal fats are examples of saturated fats.



The structure of stearic acid is shown. This fatty acid has a hydrocarbon chain seventeen residues long attached to an acetyl group. All bonds between the carbons are single bonds.

Figure 3.14 Stearic acid is a common saturated fatty acid.

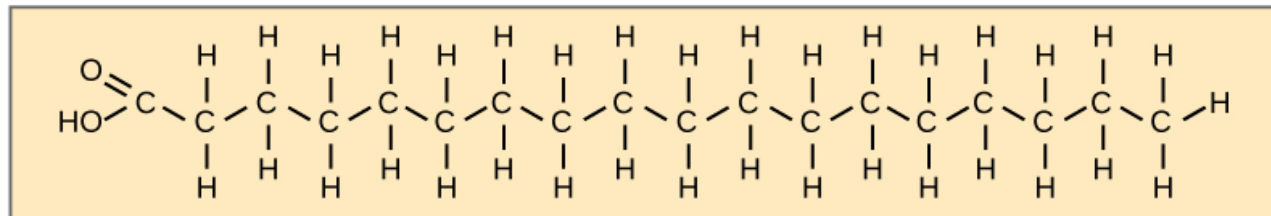
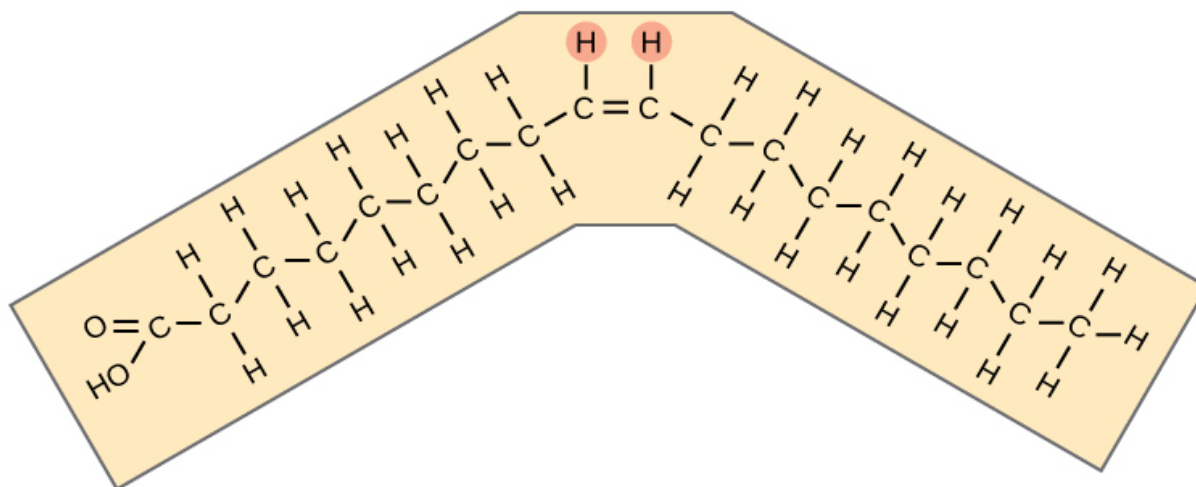
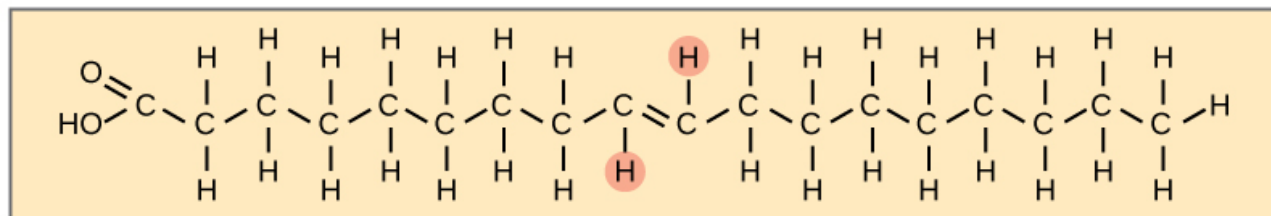
When the hydrocarbon chain contains a double bond, the fatty acid is unsaturated.



The structure of oleic acid is shown. This fatty acid has a hydrocarbon chain seventeen residues long attached to an acetyl group. The bond between carbon eight and carbon nine is a double bond.

Figure 3.15 Oleic acid is a common unsaturated fatty acid.

Most unsaturated fats are liquid at room temperature. If there is one double bond in the molecule, then it is a monounsaturated fat (e.g., olive oil), and if there is more than one double bond, then it is a polyunsaturated fat (e.g., canola oil). Unsaturated fats or oils are usually of plant origin and contain *cis* unsaturated fatty acids. *Cis* and *trans* indicate the configuration of the molecule around the double bond. The *cis* double bond causes a bend or a "kink" that prevents the fatty acids from packing tightly, keeping them liquid at room temperature ([Figure 3.16](#)).

Saturated fatty acid**Stearic acid****Unsaturated fatty acids****Cis oleic acid****Trans oleic acid**

A comparison of saturated and unsaturated fatty acids is shown. Stearic acid, a saturated fatty acid, has a hydrocarbon chain seventeen residues long attached to an acetyl group. Oleic acid also has a seventeen-residue h...

Figure 3.16 Saturated fatty acids have hydrocarbon chains connected by single bonds only. Unsaturated fatty acids have one or more double bonds. Each double bond may be in a *cis* or *trans* configuration. In the *cis* configuration, both hydrogens are on the same side of the hydrocarbon chain. In the *trans* configuration, the hydrogens are on opposite sides. A *cis* double bond causes a kink in the chain.

Trans Fats

The food industry artificially hydrogenates oils to make them semi-solid and of a consistency desirable for many processed food products. Simply speaking, hydrogen gas is bubbled through oils to solidify them. During this hydrogenation process, double bonds of the *cis*- conformation in the hydrocarbon chain may convert to double bonds in the *trans*- conformation.

Margarine, some types of peanut butter, and shortening are examples of artificially hydrogenated trans fats. Recent studies have shown that an increase in trans fats in the human diet may lead to higher levels of low-density lipoproteins (LDL), or “bad” cholesterol, which in turn may lead to plaque deposition in the arteries, resulting in heart disease. Many fast food restaurants have recently banned using trans fats, and food labels are required to display the trans fat content.

Phospholipids

Phospholipids are major plasma membrane constituents that comprise cells' outermost layer. Like fats, they are comprised of fatty acid chains attached to a glycerol backbone. However, instead of three fatty acids attached as in triglycerides, there are only two fatty acids (Figure 3.19). Phospholipids also consist of a phosphate group that is attached to the terminal end of the glycerol, and a small charged molecule attached to the terminal end of the phosphate group. This charged molecule varies in different organisms.

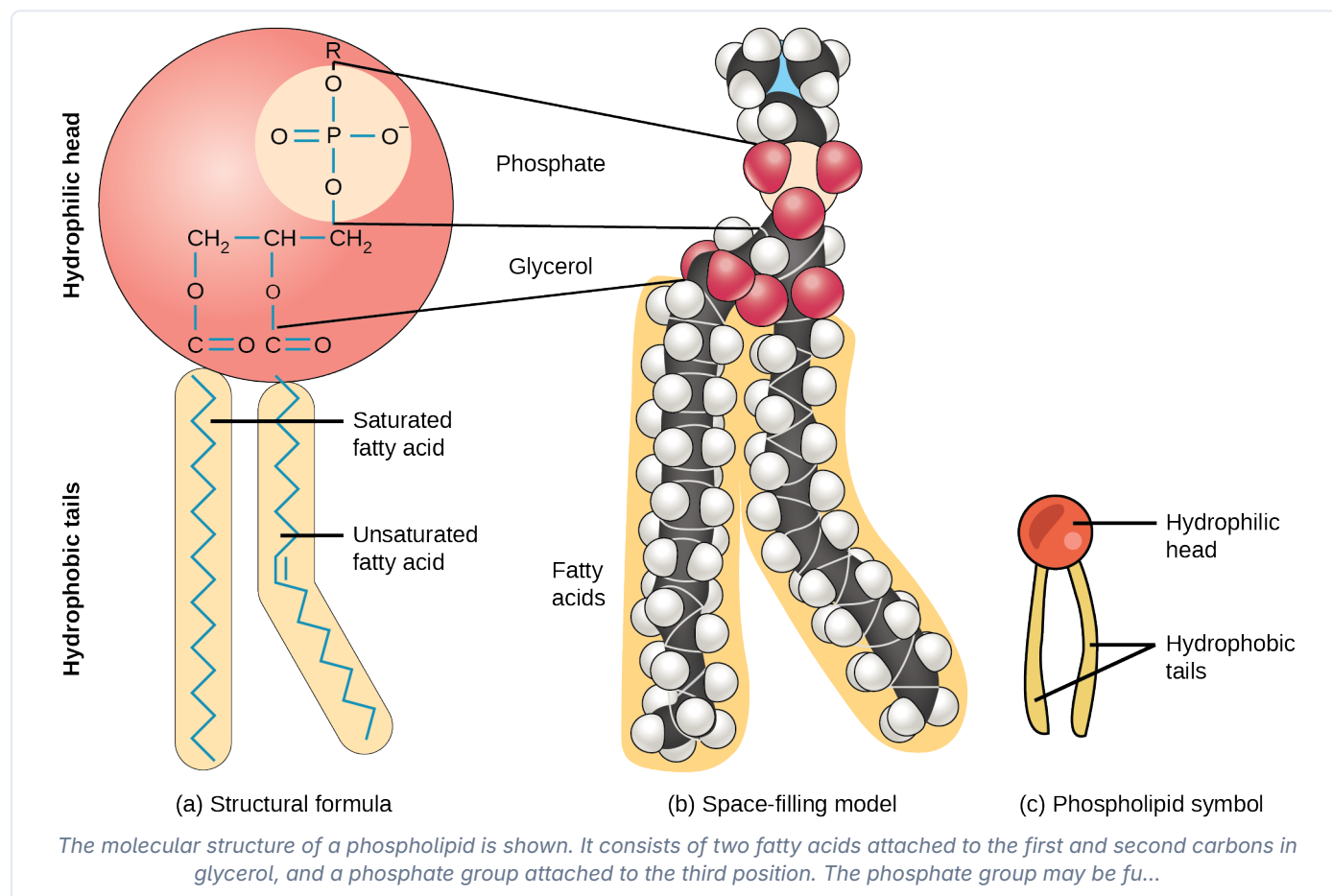
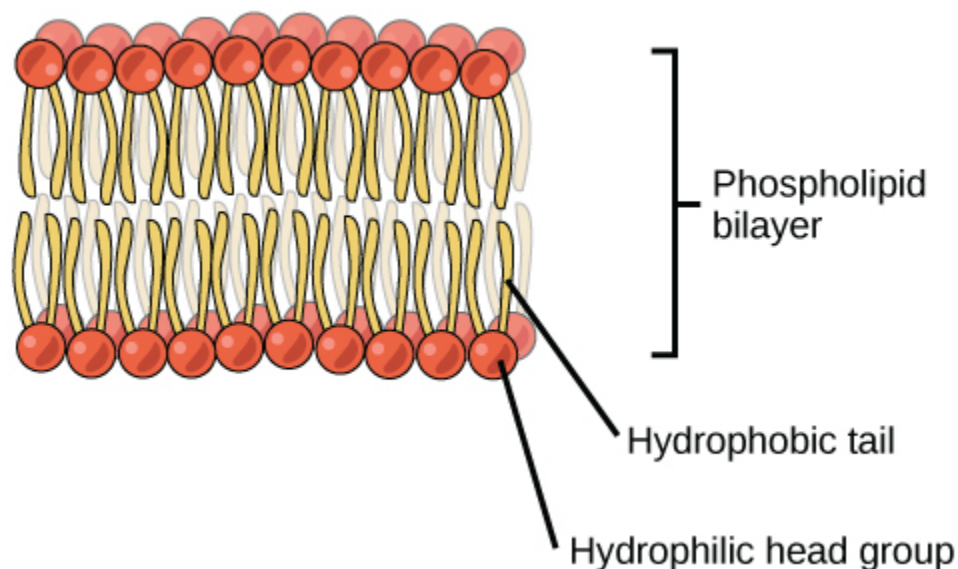


Figure 3.19 A phospholipid is a molecule with two fatty acids and a modified phosphate group attached to a glycerol backbone.

A phospholipid is an amphipathic molecule, meaning it has a hydrophobic and a hydrophilic part. The fatty acid chains are hydrophobic and cannot interact with water; whereas, the phosphate-containing group is hydrophilic and interacts with water (Figure 3.20).



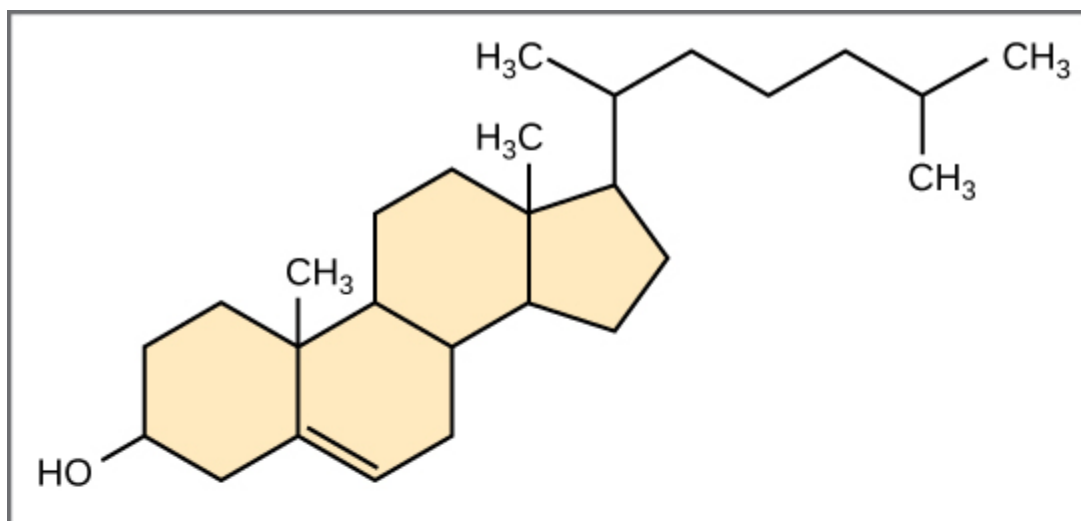
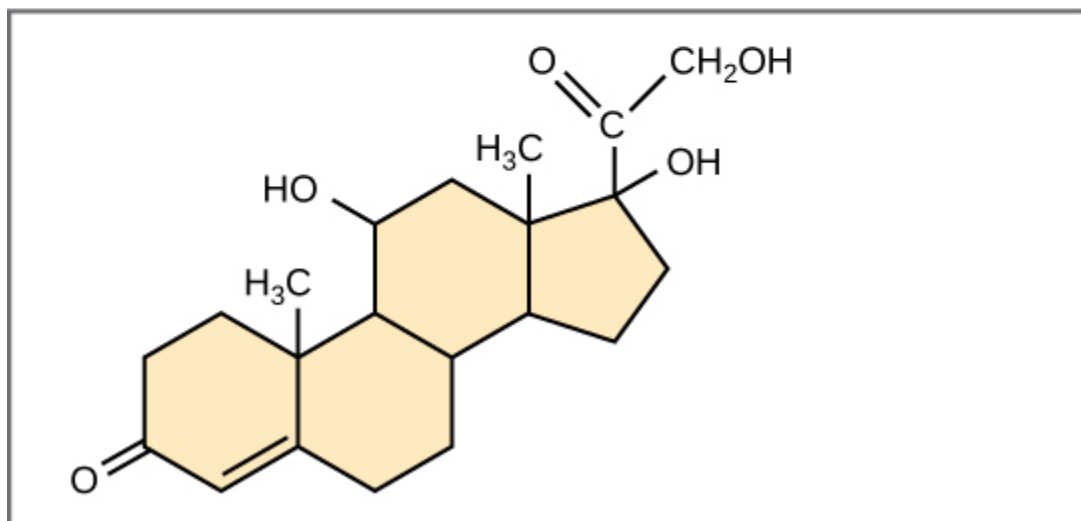
An illustration of a phospholipids bilayer is shown. The phospholipids bilayer consists of two layers of phospholipids. The hydrophobic tails of the phospholipids face one another while the hydrophilic head groups face o...

Figure 3.20 The phospholipid bilayer is the major component of all cellular membranes. The hydrophilic head groups of the phospholipids face the aqueous solution. The hydrophobic tails are sequestered in the middle of the bilayer.

The head is the hydrophilic part, and the tail contains the hydrophobic fatty acids. In a membrane, a bilayer of phospholipids forms the structure's matrix, phospholipids' fatty acid tails face inside, away from water; whereas, the phosphate group faces the outside, aqueous side (Figure 3.20). Phospholipids are responsible for the plasma membrane's dynamic nature.

Steroids

Unlike the phospholipids and fats that we discussed earlier, steroids have a fused ring structure. Although they do not resemble the other lipids, scientists group them with them because they are also hydrophobic and insoluble in water. All steroids have four linked carbon rings and several of them, like cholesterol, have a short tail (Figure 3.21).

**Cholesterol****Cortisol**

The structures of cholesterol and cortisol are shown. Each of these molecules is composed of three six-carbon rings fused to a five-carbon ring. Cholesterol has a branched hydrocarbon attached to the five-carbon ring, an...

Figure 3.21 Four fused hydrocarbon rings comprise steroids such as cholesterol and cortisol.

Cholesterol is the most common steroid. The liver synthesizes cholesterol and it is the precursor to many steroid hormones such as testosterone and estradiol, which gonads and endocrine glands secrete. It is also the precursor to Vitamin D. Cholesterol is also the precursor of bile salts, which help emulsifying fats and their subsequent absorption by cells. Although lay people often speak negatively about cholesterol, it is necessary for the body's proper functioning.

Module 3: Chapter 5 Proteins

Amino Acids, Peptide Bonds, Four Levels of Structure, and Protein Denaturation

Chapter 5 Proteins

Learning Objectives

By the end of this section, you will be able to do the following:

- Discuss the relationship between amino acids and proteins
- Explain the four levels of protein organization
- Describe the ways in which protein shape and function are linked

Proteins are one of the most abundant organic molecules in living systems and have the most diverse range of functions of all macromolecules. Proteins may be structural, regulatory, contractile, or protective. They may serve in transport, storage, or membranes; or they may be toxins or enzymes. Each cell in a living system may contain thousands of proteins, each with a unique function. Their structures, like their functions, vary greatly. They are all, however, amino acid polymers arranged in a linear sequence.

Proteins have different shapes and molecular weights. Some proteins are globular in shape; whereas, others are fibrous in nature. For example, hemoglobin is a globular protein, but collagen, located in our skin, is a fibrous protein. Protein shape is critical to its function, and many different types of chemical bonds maintain this shape. Changes in temperature, pH, and exposure to chemicals may lead to permanent changes in the protein's shape, leading to loss of function, or denaturation. Different arrangements of the same 20 types of amino acids comprise all proteins.

Amino Acids

Amino acids are the monomers that comprise proteins. Each amino acid has the same fundamental structure, which consists of a central carbon atom, called the alpha (α) carbon, bonded to an amino group (NH_2), a carboxyl group (COOH), and to a hydrogen atom. Every amino acid also has another atom or group of atoms bonded to the central atom known as the R group ([Figure 3.22](#)).

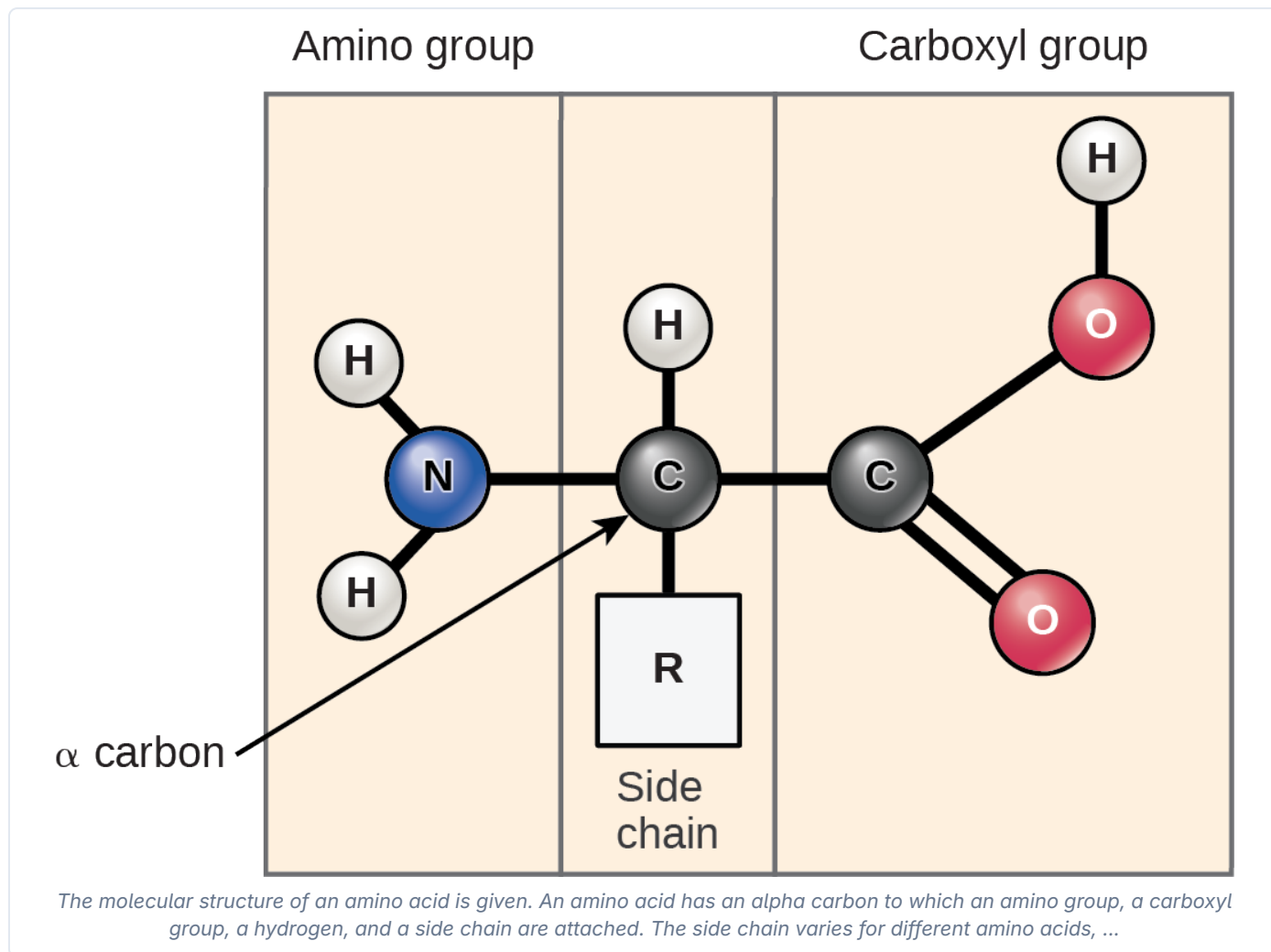
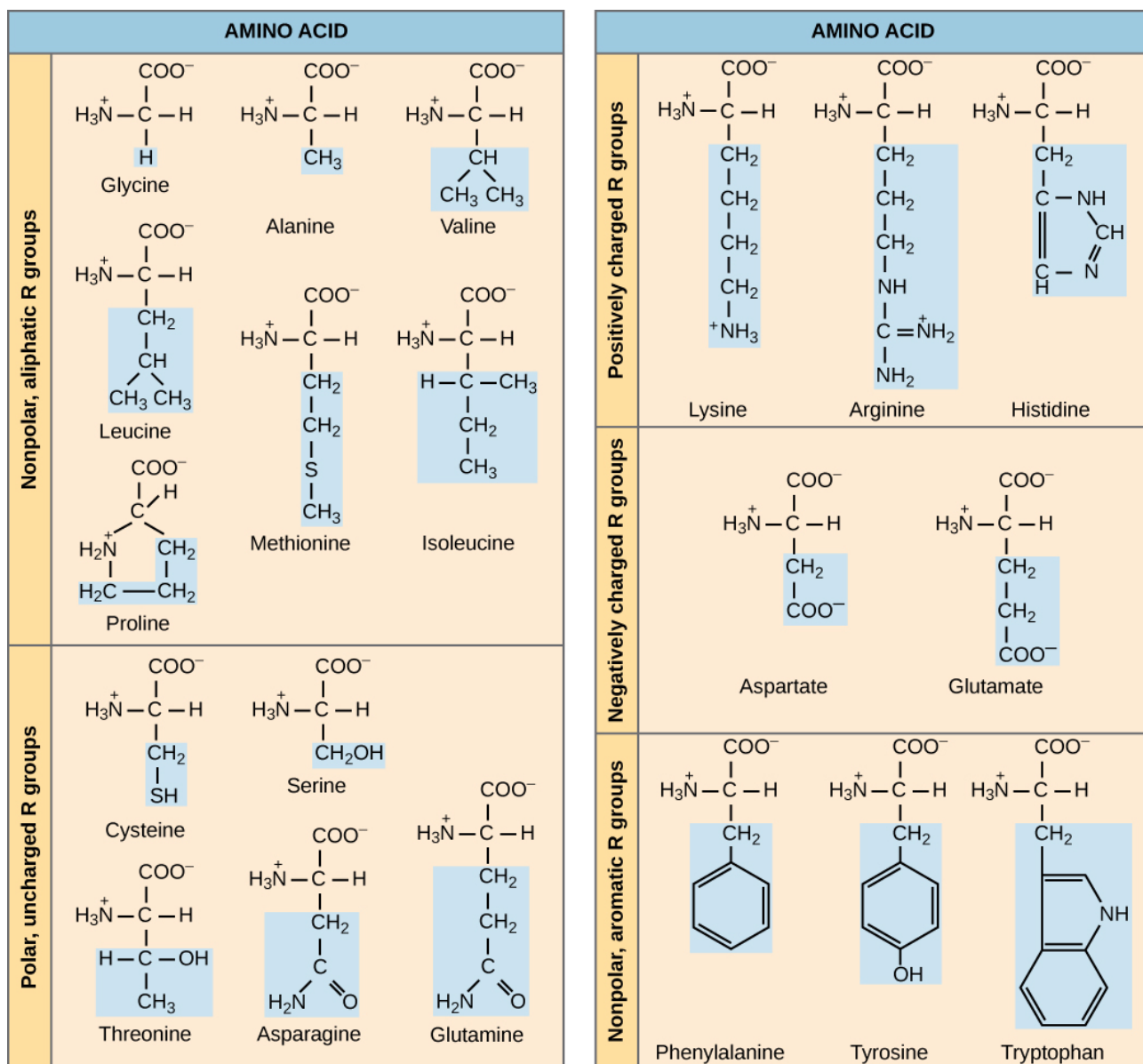


Figure 3.22 Amino acids have a central asymmetric carbon to which an amino group, a carboxyl group, a hydrogen atom, and a side chain (R group) are attached.

Scientists use the name "amino acid" because these acids contain both amino group and carboxyl-acid-group in their basic structure. As we mentioned, there are 20 common amino acids present in proteins. Nine of these are essential amino acids in humans because the human body cannot produce them and we obtain them from our diet. For each amino acid, the R group (or side chain) is different ([Figure 3.23](#)).

VISUAL CONNECTION

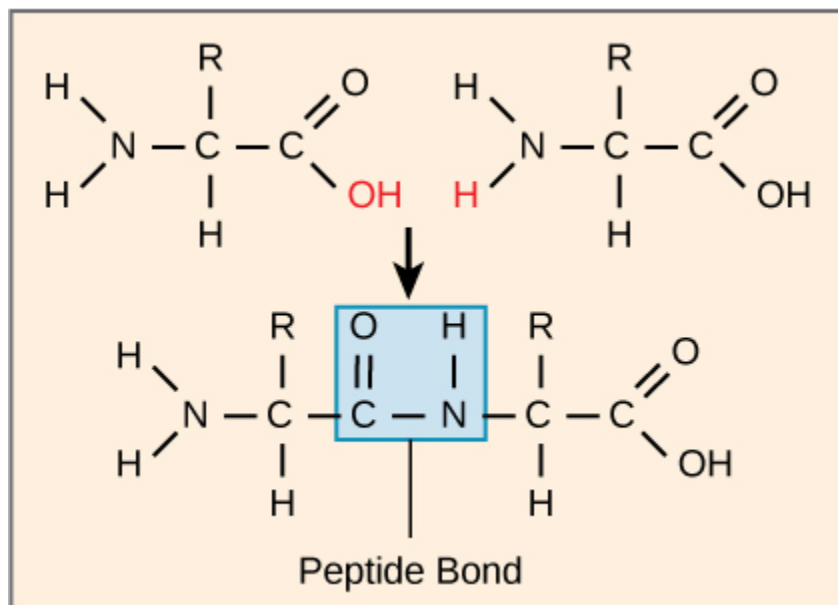
Visual Connection



The molecular structures of the twenty amino acids commonly found in proteins are given. These are divided into five categories: nonpolar aliphatic, polar uncharged, positively charged, negatively charged, and aromatic. ...

Figure 3.23 There are 20 common amino acids commonly found in proteins, each with a different R group (variant group) that determines its chemical nature.

The chemical nature of the side chain determines the amino acid's nature (that is, whether it is acidic, basic, polar, or nonpolar). A three-letter abbreviation represents amino acids. For example, the three-letter symbol "val" represent valine. The sequence and the number of amino acids ultimately determine the protein's shape, size, and function. Amino acids are linked to each other by a peptide bond. One amino acid's carboxyl group and the incoming amino acid's amino group combine, releasing a water molecule. The resulting bond is the peptide bond (Figure 3.24).



The formation of a peptide bond between two amino acids is shown. When the peptide bond forms, the carbon from the carbonyl group becomes attached to the nitrogen from the amino group. The upper O upper H from the carbox...

Figure 3.24 Peptide bond formation is a dehydration synthesis reaction. The carboxyl group of one amino acid is linked to the incoming amino acid's amino group. In the process, it releases a water molecule.

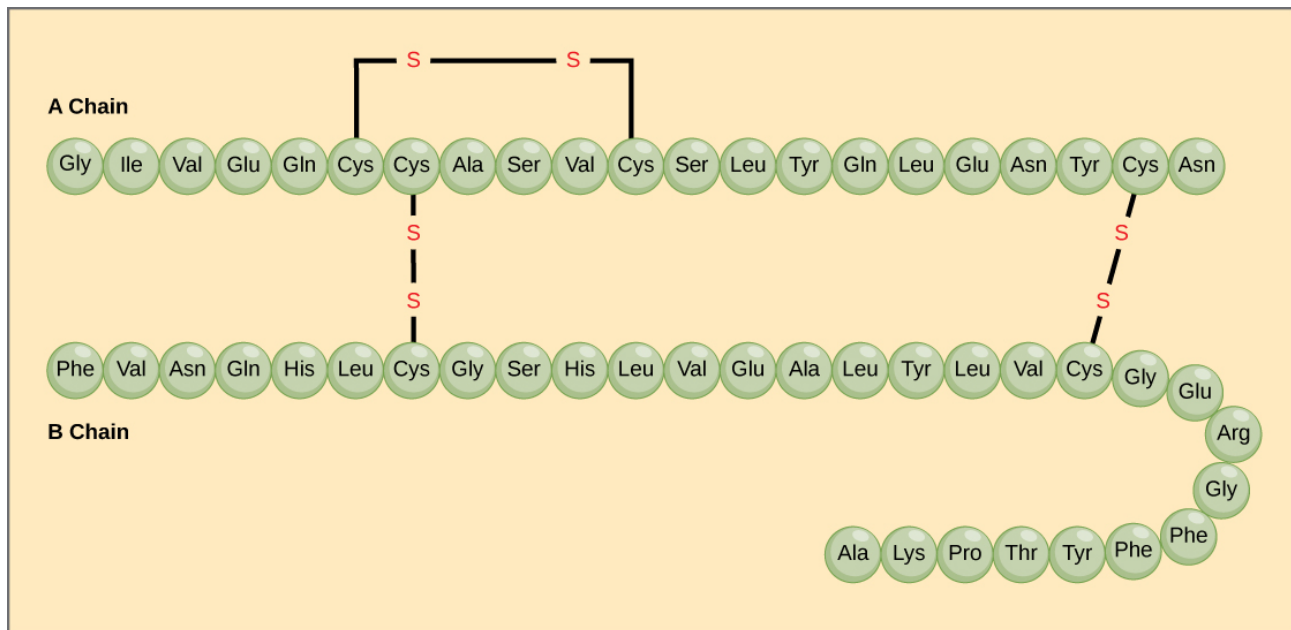
The products that such linkages form are peptides. As more amino acids join to this growing chain, the resulting chain is a polypeptide. While the terms polypeptide and protein are sometimes used interchangeably, a polypeptide is technically a polymer of amino acids linked together by peptide bonds. The term protein is reserved for a polypeptide that has undergone additional bonding, creating more complex structure. A polypeptide is not a functional protein until it has more complex structure.

Protein Structure

As we discussed earlier, a protein's shape is critical to its function. To understand how the protein gets its final shape or conformation, we need to understand the four levels of protein structure: primary, secondary, tertiary, and quaternary.

Primary Structure

Amino acids linked together by peptide bonds create a polypeptide chain, this is its **primary structure**.



The amino acid sequences for the A chain and B chain of bovine insulin are shown. The A chain is 21 amino acids in length, and the B chain is 30 amino acids in length. One disulfide, or S S bond, connects two cysteine re...

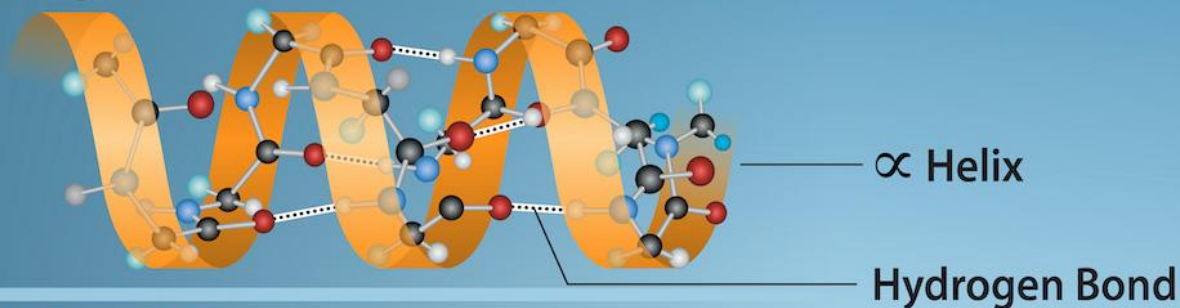
Figure 3.25 Bovine serum insulin is a protein hormone comprised of two peptide chains, A (21 amino acids long) and B (30 amino acids long). In each chain, three-letter abbreviations that represent the amino acids' names in the order they are present indicate primary structure.

Secondary Structure

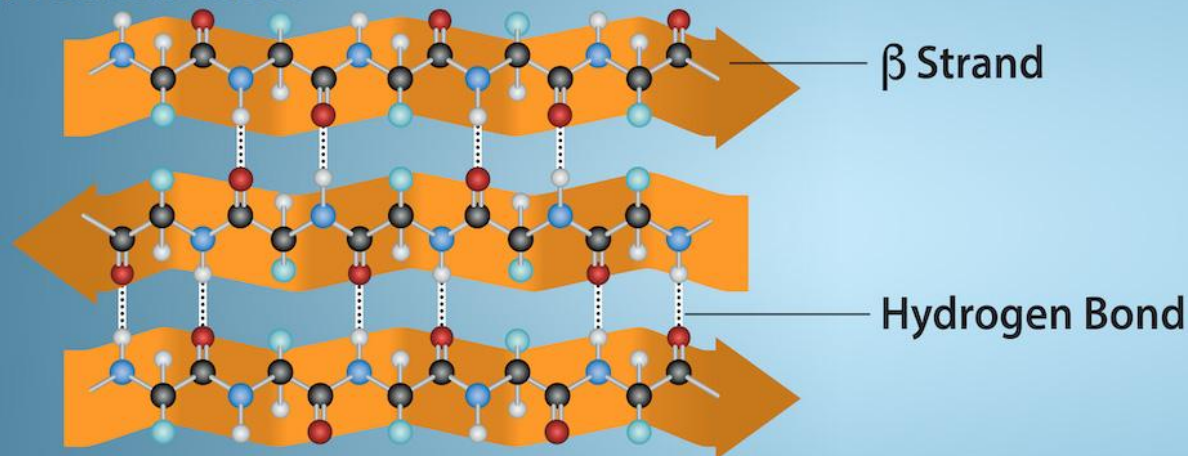
Folding of the polypeptide in some regions gives rise to the **secondary structure** of the protein. The most common are the α -helix and β -pleated sheet structures (Figure 3.28). Both structures are held in shape by **hydrogen bonds**. The hydrogen bonds form between the oxygen atom in the carbonyl group in one amino acid and another amino acid that is four amino acids farther along the chain. With the formation of α -helix and β -pleated sheet structures, the polypeptide can now be officially called a protein.

Secondary Structure

α Helix



β Pleated Sheet

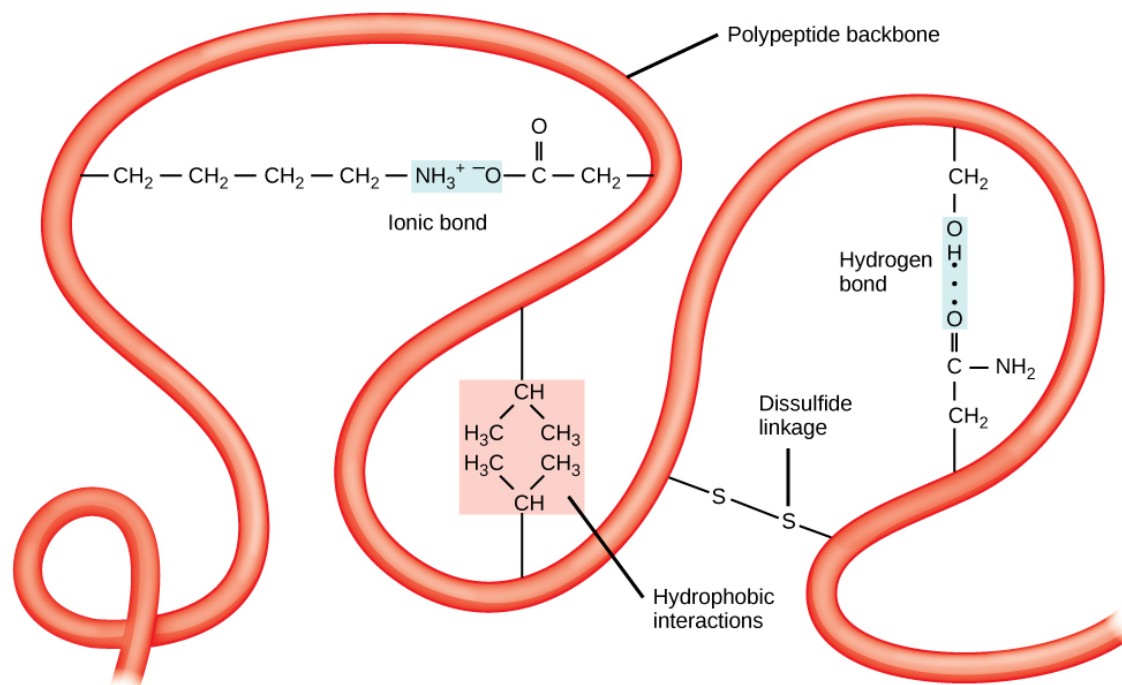


The illustration shows an alpha helix protein structure, which coils like a spring, and a beta-pleated sheet structure, which forms flat sheets stacked together. In an alpha-helix, hydrogen bonding occurs between the car...

Figure 3.28 The α -helix and β -pleated sheet are secondary protein structures formed when hydrogen bonds form between the carbonyl oxygen and the amino hydrogen in the peptide backbone. Certain amino acids have a propensity to form an α -helix while others favor β -pleated sheet formation. Black = carbon, White = hydrogen, Blue = nitrogen, and Red = oxygen. Credit: Rao, A., Ryan, K. Fletcher, S. and Tag, A. Department of Biology, Texas A&M University.

Tertiary Structure

The polypeptide's unique three-dimensional structure is its **tertiary structure** (Figure 3.29). This structure is due to interactions among R groups of individual amino acids. The nature of the R groups in the amino acids ultimately determine the complex three-dimensional **tertiary structure**. For example, R groups with like charges repel each other and those with unlike charges are attracted to each other (ionic bonds). When protein folding takes place, the nonpolar amino acids' hydrophobic R groups lie in the protein's interior; whereas, the hydrophilic R groups lie on the outside.



This illustration shows a polypeptide backbone folded into a three-dimensional structure. Chemical interactions between amino acid side chains maintain its shape. These include an ionic bond between an amino group and a ...

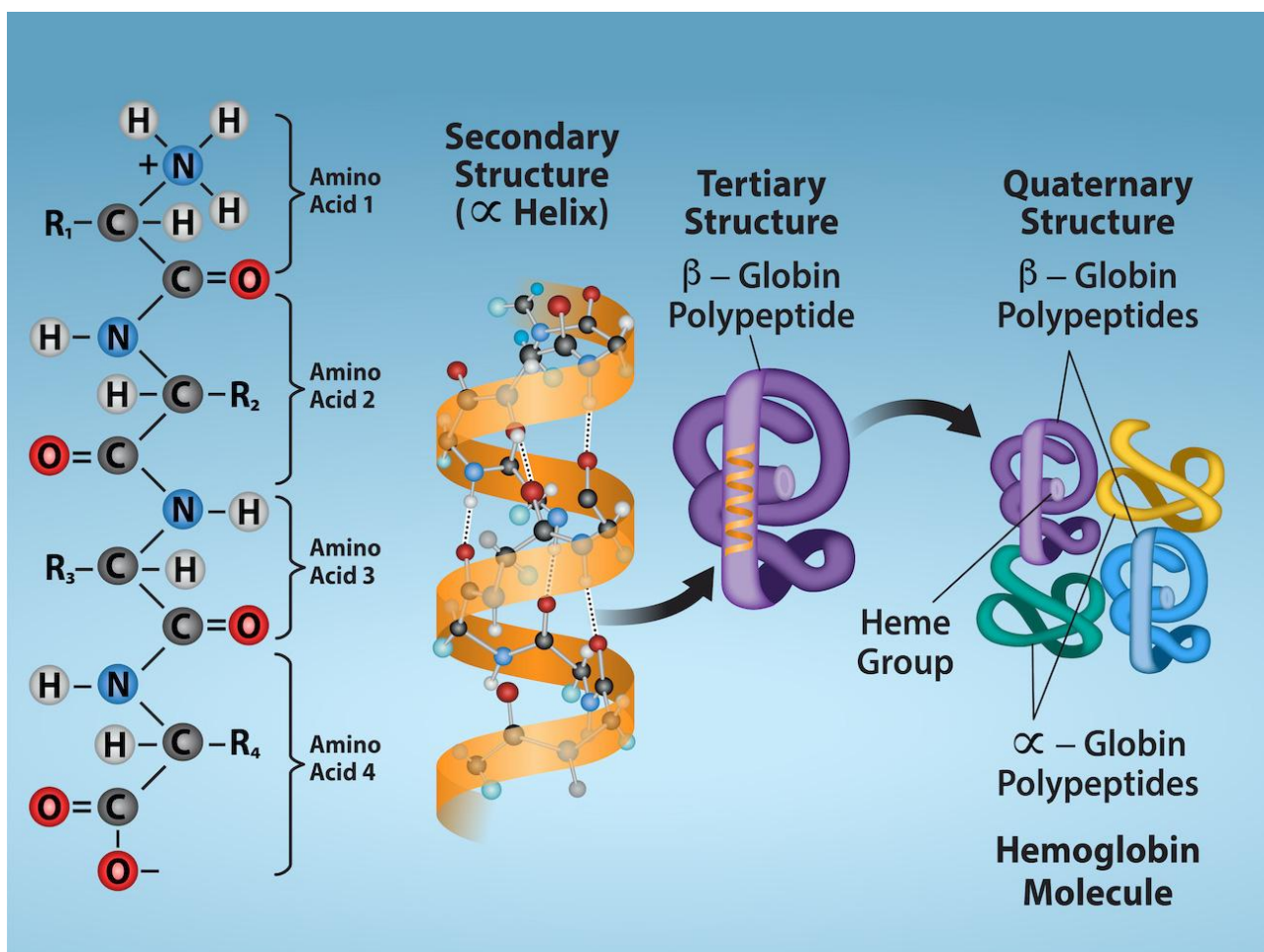
Figure 3.29 A variety of chemical interactions determine the proteins' tertiary structure. These include hydrophobic interactions, ionic bonding, hydrogen bonding, and disulfide linkages.

All of these interactions, weak and strong, determine the protein's final three-dimensional shape. If a protein loses its three-dimensional shape, it may no longer be functional.

Quaternary Structure

In nature, some proteins form aggregates combining several smaller proteins into one large, very complex protein. These aggregate proteins form the **quaternary structure**.

Figure 3.30 illustrates the four levels of protein structure (primary, secondary, tertiary, and quaternary).



Shown are the four levels of protein structure. The primary structure is the amino acid sequence. Secondary structure is a regular folding pattern due to hydrogen bonding. Tertiary structure is the three-dimensional fold...

Figure 3.30 Observe the four levels of protein structure in these illustrations. Credit: Rao, A. Ryan, K. and Tag, A. Department of Biology, Texas A&M University.

Denaturation and Protein Folding

Each protein has its own unique sequence and shape that chemical interactions hold together. If the protein is subject to changes in temperature, pH, or exposure to chemicals, the protein structure may change, losing its shape without losing its primary structure. This is called **denaturation**. Denaturation is often reversible because the polypeptide's primary structure is conserved in the process if the denaturing agent is removed, allowing the protein to resume its function. Sometimes denaturation is irreversible, leading to loss of function. One example of irreversible protein denaturation is frying an egg. The albumin protein in the liquid egg white denatures when placed in a hot pan.

Protein folding is critical to its function. Scientists originally thought that the proteins themselves were responsible for the folding process. Only recently researchers discovered that often they receive assistance in the folding process from protein helpers, called chaperonins.

Module 4: Chapter 5 Nucleic Acids

Nucleotides, Phosphodiester Bonds, DNA Double Helix, and RNA Types

Chapter 5 Nucleic Acids

Learning Objectives

By the end of this section, you will be able to do the following:

- Describe nucleic acids' structure and define the two types of nucleic acids
- Explain DNA's structure and role
- Explain RNA's structure and roles

Nucleic acids are the most important macromolecules for the continuity of life. They carry the cell's genetic blueprint and carry instructions for its functioning.

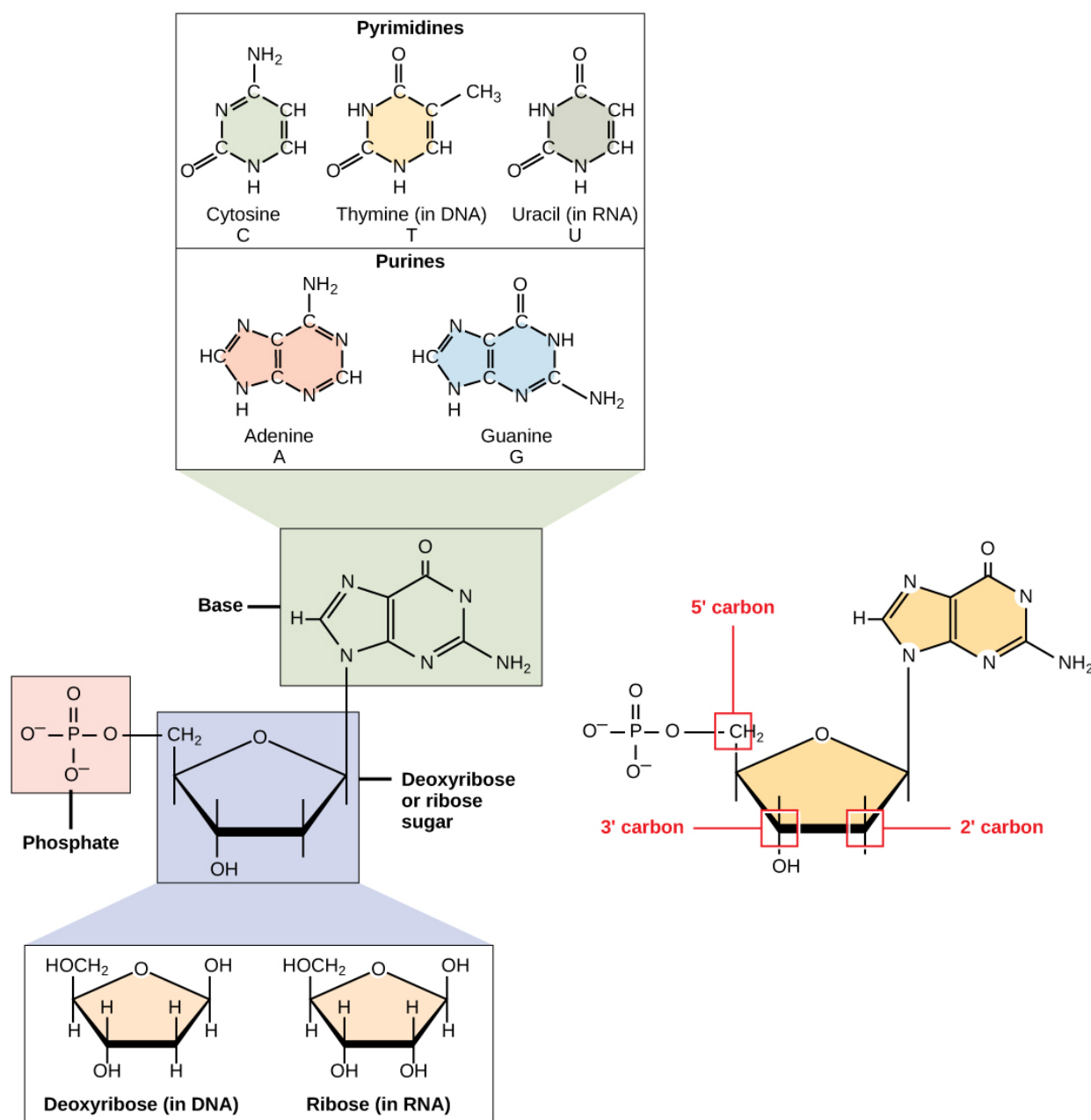
DNA and RNA

The two main types of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). DNA is the genetic material in all living organisms, ranging from single-celled bacteria to multicellular mammals. It is in the nucleus of eukaryotes and in the organelles, chloroplasts, and mitochondria. In prokaryotes, the DNA is not enclosed in a membranous envelope.

The cell's entire genetic content is its genome, and the study of genomes is called genomics. In eukaryotic cells, but not in prokaryotes, DNA forms a complex with histone proteins to form chromatin, the substance of eukaryotic chromosomes. A chromosome may contain tens of thousands of genes. Many genes contain the information to make protein products. Other genes code for RNA products. DNA controls all of the cellular activities by turning the genes "on" or "off."

The other type of nucleic acid, RNA, is mostly involved in protein synthesis. The DNA molecules never leave the nucleus but instead use an intermediary to communicate with the rest of the cell. This intermediary is the messenger RNA (mRNA). Other types of RNA—like rRNA, tRNA, and microRNA—are involved in protein synthesis and its regulation.

DNA and RNA are comprised of monomers that scientists call nucleotides. The nucleotides combine with each other to form a polynucleotide, DNA or RNA. Three components comprise each nucleotide: a nitrogenous base, a pentose (five-carbon) sugar, and a phosphate group ([Figure 3.31](#)). Each nitrogenous base in a nucleotide is attached to a sugar molecule, which is attached to one or more phosphate groups.



The molecular structure of a nucleotide is shown. The core of the nucleotide is a pentose whose carbon residues are numbered one prime through five prime. The base is attached to the one prime carbon, and the phosphate i...

Figure 3.31 Three components comprise a nucleotide: a nitrogenous base, a pentose sugar, and one or more phosphate groups. Carbon residues in the pentose are numbered 1' through 5' (the prime distinguishes these residues from those in the base, which are numbered without using a prime notation). The base is attached to the ribose's 1' position, and the phosphate is attached to the 5' position. Two types of pentose are in nucleotides, deoxyribose (found in DNA) and ribose (found in RNA). Deoxyribose is similar in structure to ribose, but it has an H instead of an OH at the 2' position. We can divide bases into two categories: purines and pyrimidines. Purines have a double ring structure, and pyrimidines have a single ring.

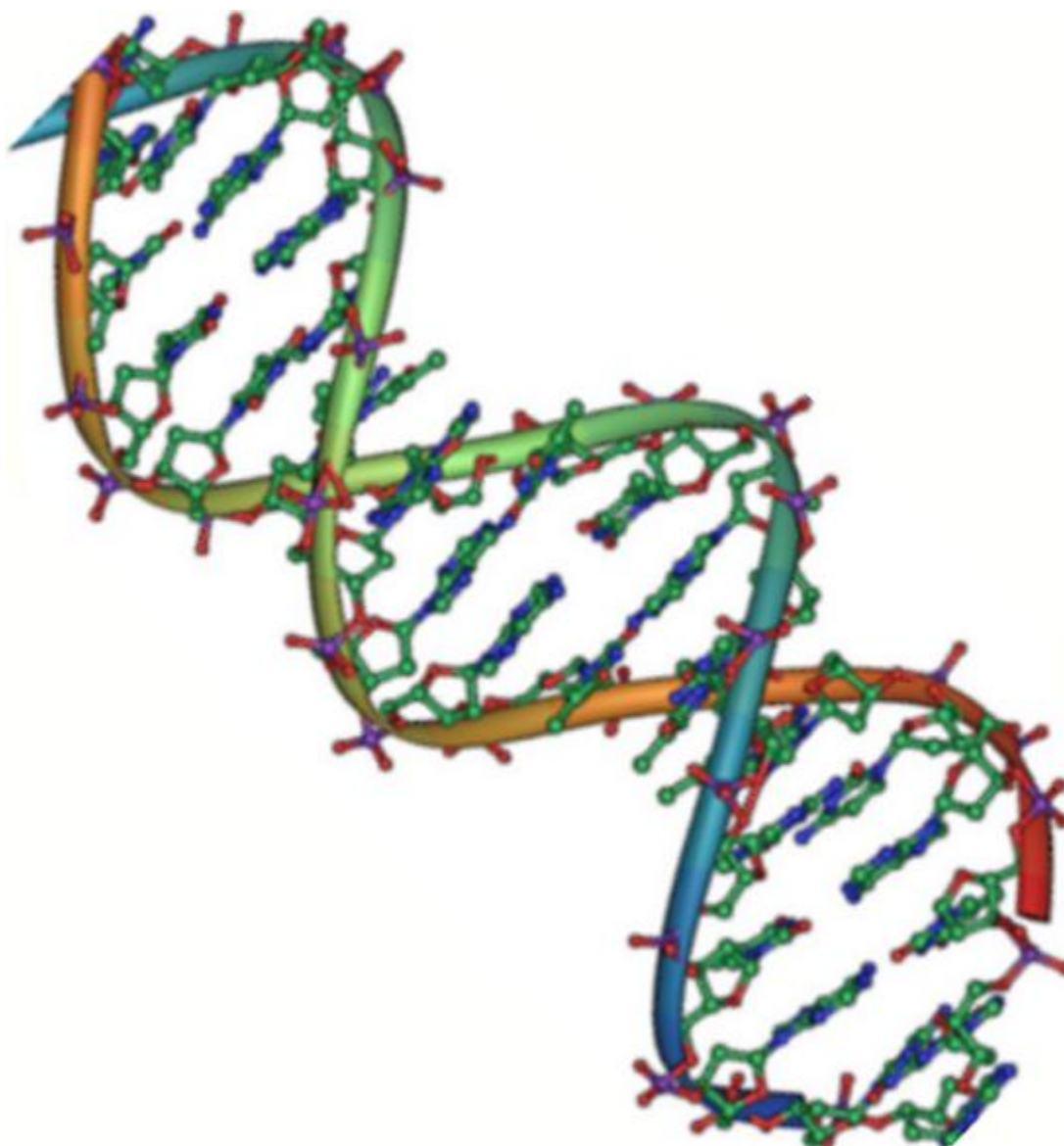
The nitrogenous bases, important components of nucleotides, are organic molecules and are so named because they contain carbon and nitrogen. Each nucleotide in DNA contains one of four possible nitrogenous bases: adenine (A), guanine (G) cytosine (C), and thymine (T).

Scientists classify adenine and guanine as purines. The purine's primary structure is two carbon-nitrogen rings. Scientists classify cytosine, thymine, and uracil as pyrimidines which have a single carbon-nitrogen ring as their primary structure (Figure 3.31). In molecular biology shorthand, we know the nitrogenous bases by their symbols A, T, G, C, and U. DNA contains A, T, G, and C; whereas, RNA contains A, U, G, and C.

The pentose sugar in DNA is deoxyribose, and in RNA, the sugar is ribose (Figure 3.31). The difference between the sugars is the presence of the hydroxyl group on the ribose's second carbon and hydrogen on the deoxyribose's second carbon. The carbon atoms of the sugar molecule are numbered as 1', 2', 3', 4', and 5' (1' is read as "one prime"). The phosphate residue attaches to the hydroxyl group of the 5' carbon of one sugar and the hydroxyl group of the 3' carbon of the sugar of the next nucleotide, which forms a 5'–3' **phosphodiester linkage**.

DNA Double-Helix Structure

DNA has a double-helix structure (Figure 3.32). The sugar and phosphate lie on the outside of the helix, forming the DNA's backbone. The nitrogenous bases are stacked in the interior, like a pair of staircase steps. Hydrogen bonds bind the pairs to each other. The helix's two strands run in opposite directions, meaning that the 5' carbon end of one strand will face the 3' carbon end of its matching strand. (Scientists call this an antiparallel orientation and is important to DNA replication and in many nucleic acid interactions.)



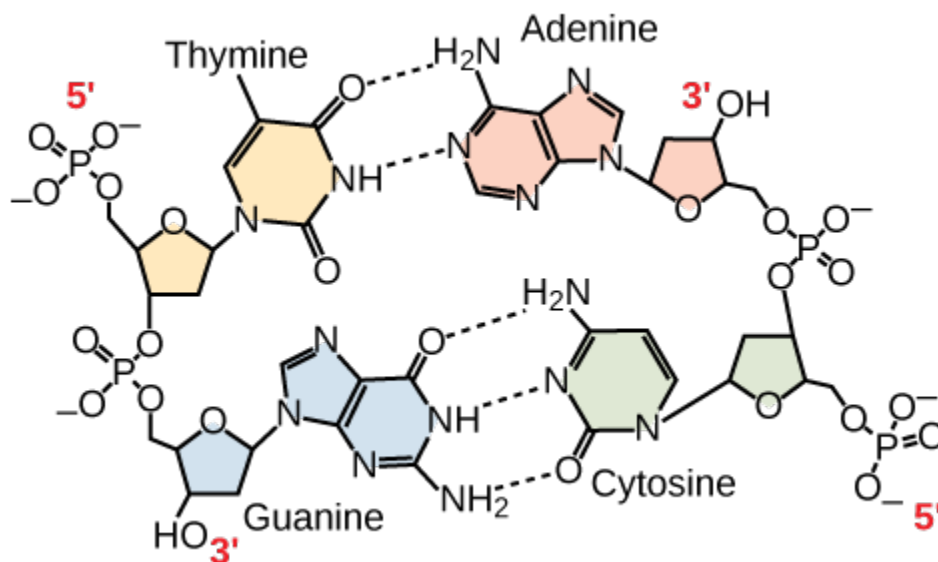
The molecular structure of D N A is shown. D N A consists of two antiparallel strands twisted in a double helix. The phosphate backbone is on the outside, and the nitrogenous bases face one another on the inside. The ba...

Figure 3.32 Native DNA is an antiparallel double helix. The phosphate backbone (indicated by the curvy lines) is on the outside, and the bases are on the inside. Each base from one strand interacts via hydrogen bonding with a base from the opposing strand. (credit: Jerome Walker/Dennis Myts)

Only certain types of base pairing are allowed. For example, a certain purine can only pair with a certain pyrimidine. This means A can pair with T, and G can pair with C, as [Figure 3.33](#) shows. This is the base complementary rule. In other words, the DNA strands are complementary to each other. If the sequence of one strand is AATTGGCC, the complementary strand would have the sequence TTAACCGG. During DNA replication, each strand copies itself, resulting in a daughter DNA double helix containing one parental DNA strand and a newly synthesized strand.

VISUAL CONNECTION

Visual Connection



Hydrogen bonding between thymine and adenine and between guanine and cytosine is shown. Thymine forms two hydrogen bonds with adenine, and guanine forms three hydrogen bonds with cytosine. The phosphate backbones of each...

Figure 3.33 In a double stranded DNA molecule, the two strands run antiparallel to one another so that one strand runs 5' to 3' and the other 3' to 5'. The phosphate backbone is located on the outside, and the bases are in the middle. Adenine forms hydrogen bonds (or base pairs) with thymine, and guanine base pairs with cytosine.

RNA

Ribonucleic acid, or RNA, is mainly involved in the process of protein synthesis under the direction of DNA. RNA is usually single-stranded and is comprised of ribonucleotides that are linked by phosphodiester bonds. A ribonucleotide in the RNA chain contains ribose (the pentose sugar), one of the four nitrogenous bases (A, U, G, and C), and the phosphate group.

DNA and RNA Features

	DNA	RNA
Function	Carries genetic information	Involved in protein synthesis
Location	Remains in the nucleus	Leaves the nucleus
Structure	Double helix	Usually single-stranded
Sugar	Deoxyribose	Ribose
Pyrimidines	Cytosine, thymine	Cytosine, uracil
Purines	Adenine, guanine	Adenine, guanine

Table 3.2

Even though the RNA is single stranded, most RNA types show extensive intramolecular base pairing between complementary sequences, creating a predictable three-dimensional structure essential for their function.